Experiment No: 04

Title: Implement Database Management using arrays of structures with operations Create, Display, Modify,Search and Sort.

Theory:

A constructed data type know as structure which is method for packing data of different types is convenient tool for handling a group of logically related items. A structure is a collection of data fields or variables of different types that is referenced under the same name. It provides convenience means of keeping related information which are called fields of members can be accessed and processed separately.

Code:

```
#include<stdio.h>

struct employee{
   int id;
   char name[50];
   float salary;
};

int main(){
   struct employee e[100],temp;
   int n=0,i,j,choice,id,found;

   while(1){
      printf("\n\n1. Create");
      printf("\n2. Display");
      printf("\n3. Modify");
      printf("\n4. Append");
      printf("\n5. Search");
      printf("\n6. Sort");
      printf("\n7. Exit");

      printf("\nEnter your choice: ");
      scanf("%d",&choice);

      switch(choice){

         case 1:
            printf("Enter number of employees: ");
            scanf("%d",&n);

            for(i=0;i<n;i++){
               printf("\nEnter ID, Name and Salary: ");
               scanf("%d%s%f",&e[i].id,e[i].name,&e[i].salary);
```

```
    }
    break;

case 2:
    printf("\nID\tName\tSalary\n");
    for(i=0;i<n;i++){
        printf("%d\t%s\t%.2f\n",e[i].id,e[i].name,e[i].salary);
    }
    break;

case 3:
    printf("Enter employee ID to modify: ");
    scanf("%d",&id);
    found=0;

    for(i=0;i<n;i++){
        if(e[i].id==id){
            printf("Enter new Name and Salary: ");
            scanf("%s%f",e[i].name,&e[i].salary);
            found=1;
            break;
        }
    }

    if(found==0)
        printf("Employee not found");
    break;

case 4:
    if(n<100){
        printf("Enter ID, Name and Salary: ");
        scanf("%d%s%f",&e[n].id,e[n].name,&e[n].salary);
        n++;
    }
    break;

case 5:
    printf("Enter employee ID to search: ");
    scanf("%d",&id);
    found=0;

    for(i=0;i<n;i++){
        if(e[i].id==id){
            printf("ID: %d\nName: %s\nSalary: %.2f",
            e[i].id,e[i].name,e[i].salary);
            found=1;
            break;
        }
    }
```

```
            if(found==0)
                printf("Employee not found");
            break;

        case 6:
            for(i=0;i<n-1;i++){
                for(j=0;j<n-i-1;j++){
                    if(e[j].id>e[j+1].id){
                        temp=e[j];
                        e[j]=e[j+1];
                        e[j+1]=temp;
                    }
                }
            }
            printf("Database sorted by employee ID");
            break;

        case 7:
            return 0;

        default:
            printf("Invalid choice");
      }
   }

   return 0;
}
```

Output:
1. Create:
Enter number of employees: 2
101 Rahul 25000
102 Amit 30000

2. Display:
101 Rahul 25000.00
102 Amit   30000.00

3. Modify:
Enter ID: 101
New Name and Salary: Raj 28000

4. Append:
Enter ID, Name and Salary: 103 Neha 35000

5. Search:
Enter ID: 102
ID: 102

Name: Amit
Salary: 30000.00

6. Sort:
Database sorted by employee ID.

7. Exit:
Program terminated.

M.M.C. O. E.

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Assignment 4.

Q1 Define data, data item & entity.

-> Data are simply values or sets of values. Data item is single unit of values. entity is something which has certain attributes which may be assigned values.

Q2 Define Data structure.

-> A data structure is a way of organizing data that considers not only the items stored but also their relationship to each other. Advance knowledge about the rln b/w data items allows designing of efficient algorithms for manipulation of data.

Q3 What are the types of data str?

-> linear & non linear.
array, linked list & trees & graphs.

Q4 What is nested structure?

-> Nested structure in C is nothing but str within structure

Q5 What are Unions?

-> Unions are same as structures. Unions are constructed/user defined data type which can grp together diff data types.

Q6 Compare structure with Unions.

-> All members of structures can be accessed at any time. But only one member of Union can be accessed at a time in case of Union & other member will contain garbage value.